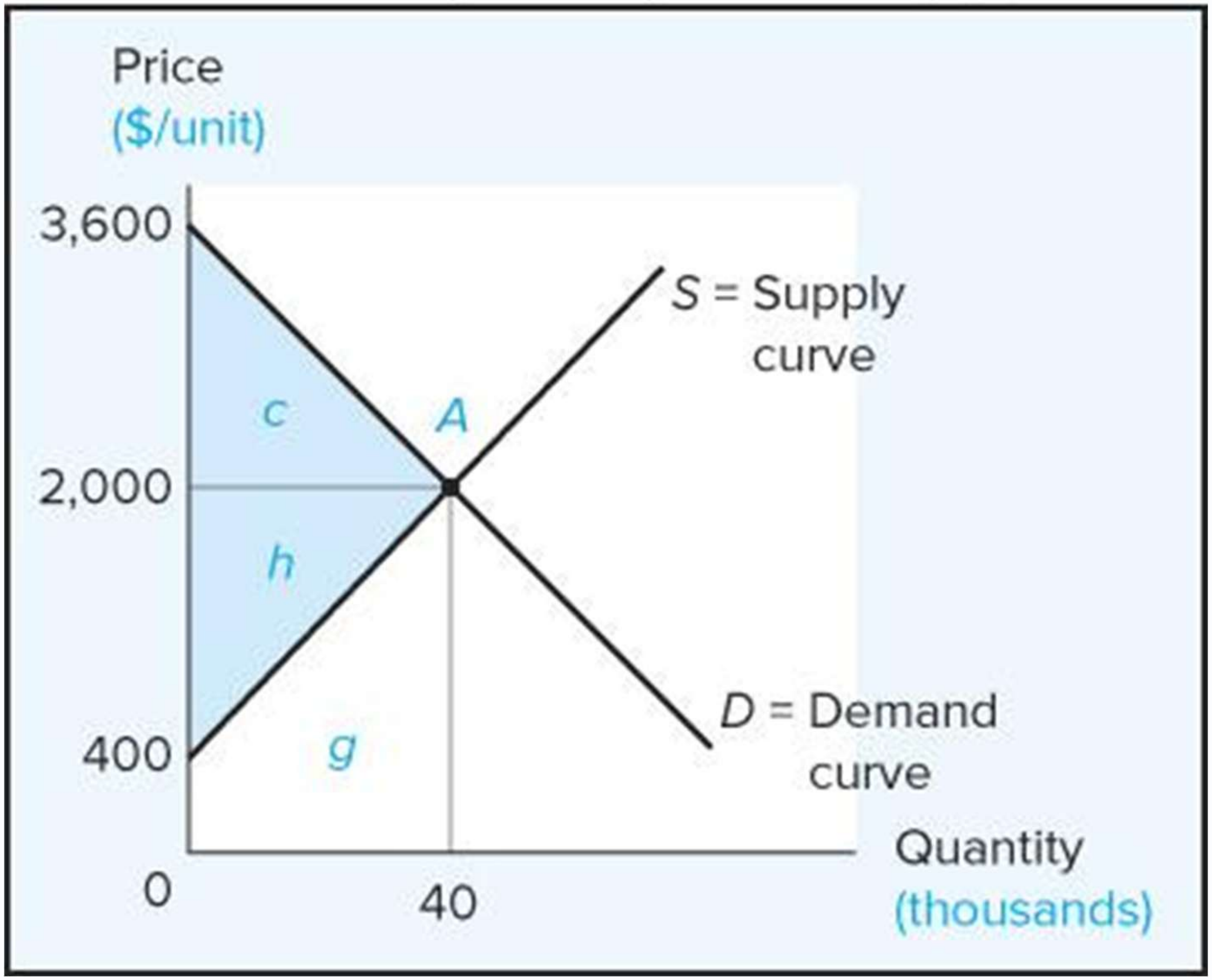
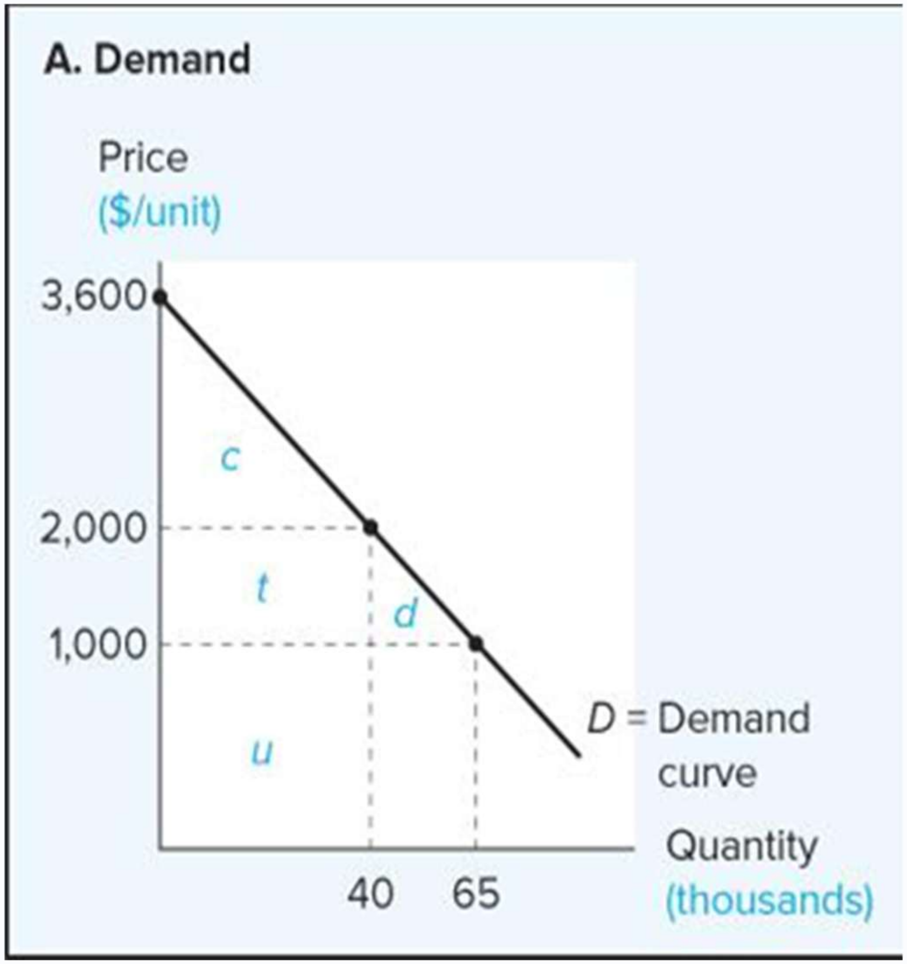
Chapter 2: The Basic Theory Using Demand and Supply

International Economics, 18th Edition, Pugel

The Market for Motorbikes: Demand and Supply



Demand Curve



Shows the quantity consumers will buy at each possible market price, other things equal.

$$P = a - bQ_d$$

$$a = 3{,}600$$

$$b = -\frac{1{,}000}{65k - 40k} Q_d$$

$$\boldsymbol{P = 3600 - 0.04\, Q_d}$$

Supply Curve

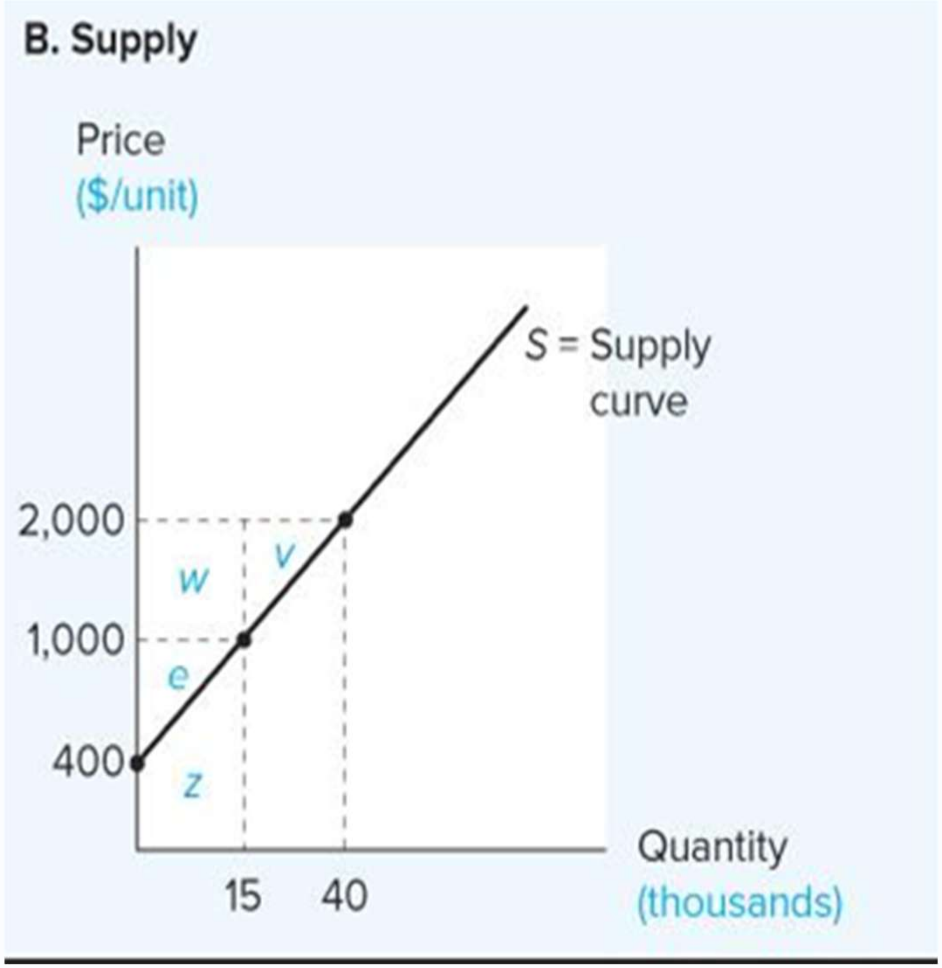
Shows the quantity that producers will offer for sale at each possible market price, other things equal.

$$P = a + bQ_s$$

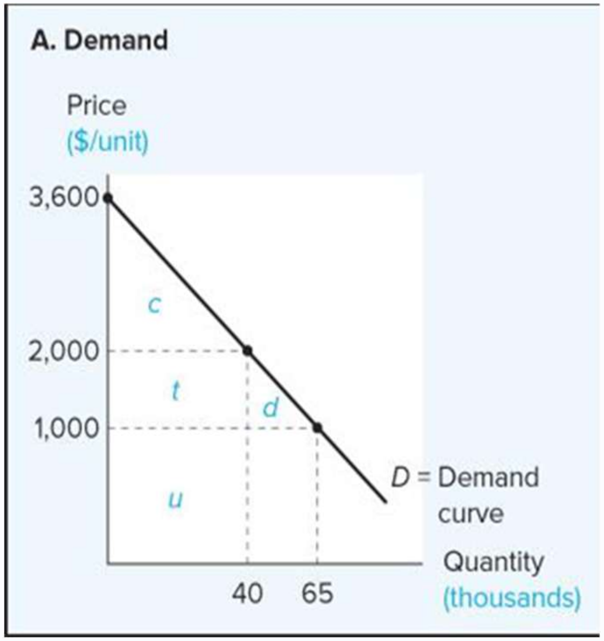
$$a = 400$$

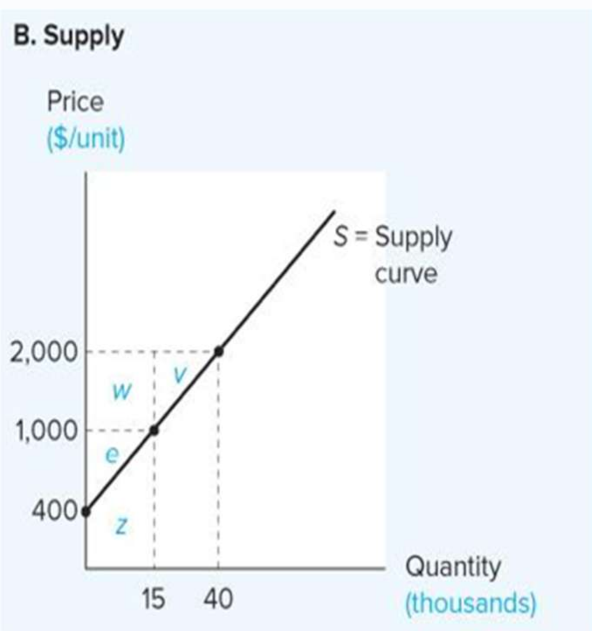
$$b = \frac{1000}{(40k - 15k)}$$

$$\boldsymbol{P = 400 + 0.04\, Q_s}$$



Simultaneous Equations
Calculate the intersection of Demand & Supply





$$P = 3600 - 0.04\,Q_d \quad \& \quad P = 400 + 0.04\,Q_s$$

$$3600 - 0.04\,Q_d = 400 + 0.04\,Q_s$$

$$3600 - 0.04\,Q \quad - 400 = 400 + 0.04\,Q \quad - 400$$

$$3200 - 0.04\,Q \quad + 0.04Q \quad = 0.04\,Q \quad + 0.04Q$$

$$\frac{3200}{0.08} = \frac{0.08\,Q}{0.08}$$

$$Q = \mathbf{40,000}$$

$$P = 3600 - 0.04\,x\,40{,}000 = \$\mathbf{2,000}$$

Supply & Demand Elasticity

Elasticity is a measure of responsiveness that is "unit-free"

Price elasticity of demand

% change in quantity demanded resulting from a 1% price change.

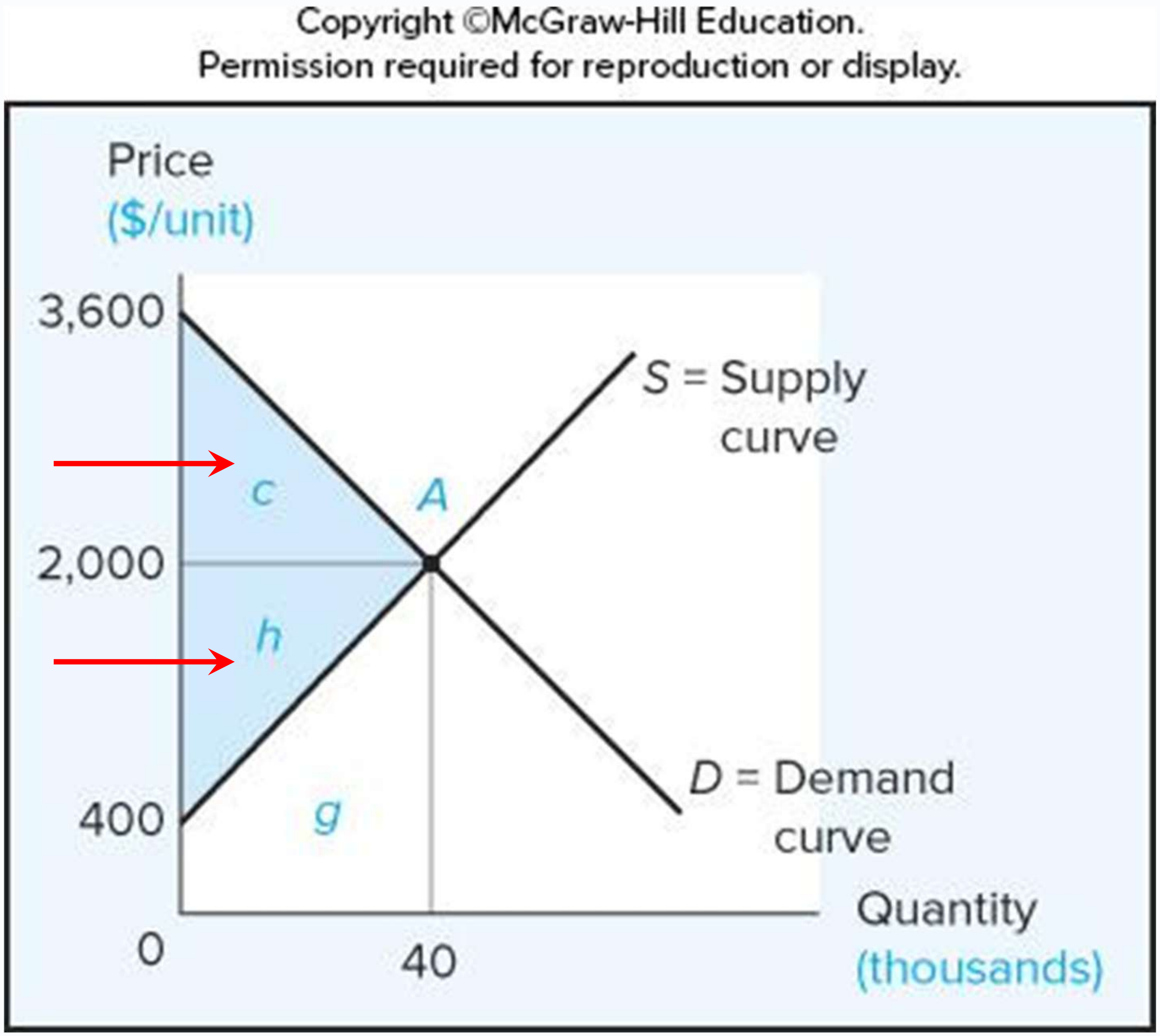
Price elasticity of supply

% increase in quantity supplied resulting from a 1% increase in price.

The Market for Motorbikes: Demand and Supply

consumer surplus

producer surplus



Two National Markets and the Opening of Trade

Arbitrage is buying something in one market and reselling the same thing in another market to profit from a price difference.

With no trade, differences in the price of a product between national markets creates an arbitrage opportunity that can lead to international trade.

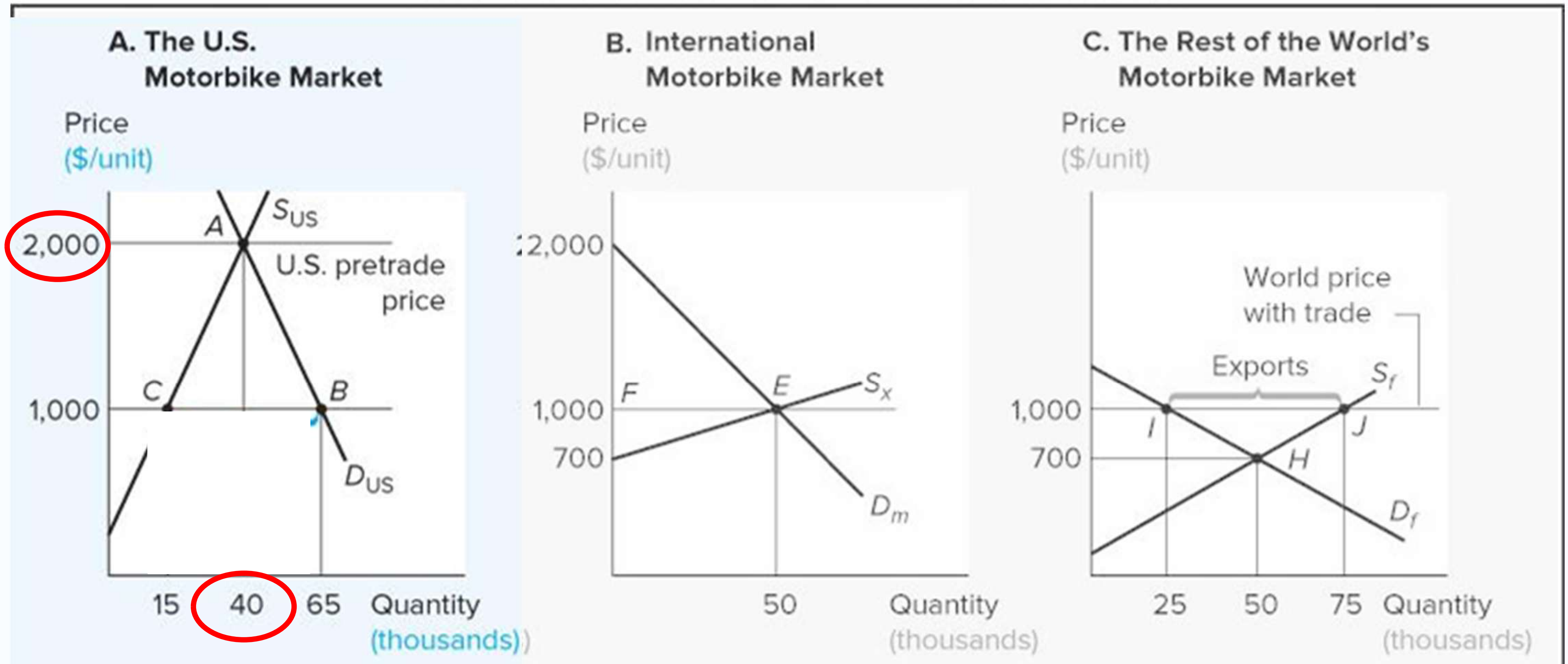
Retail Arbitrage

Part 2: Free Trade Equilibrium

Free Trade Equilibrium

As international trade in a product (e.g., motorbikes) develops between the two countries, it affects the equilibrium price in importing and exporting countries.

The Effects of Trade on Production, Consumption, and Price with Demand and Supply Curves



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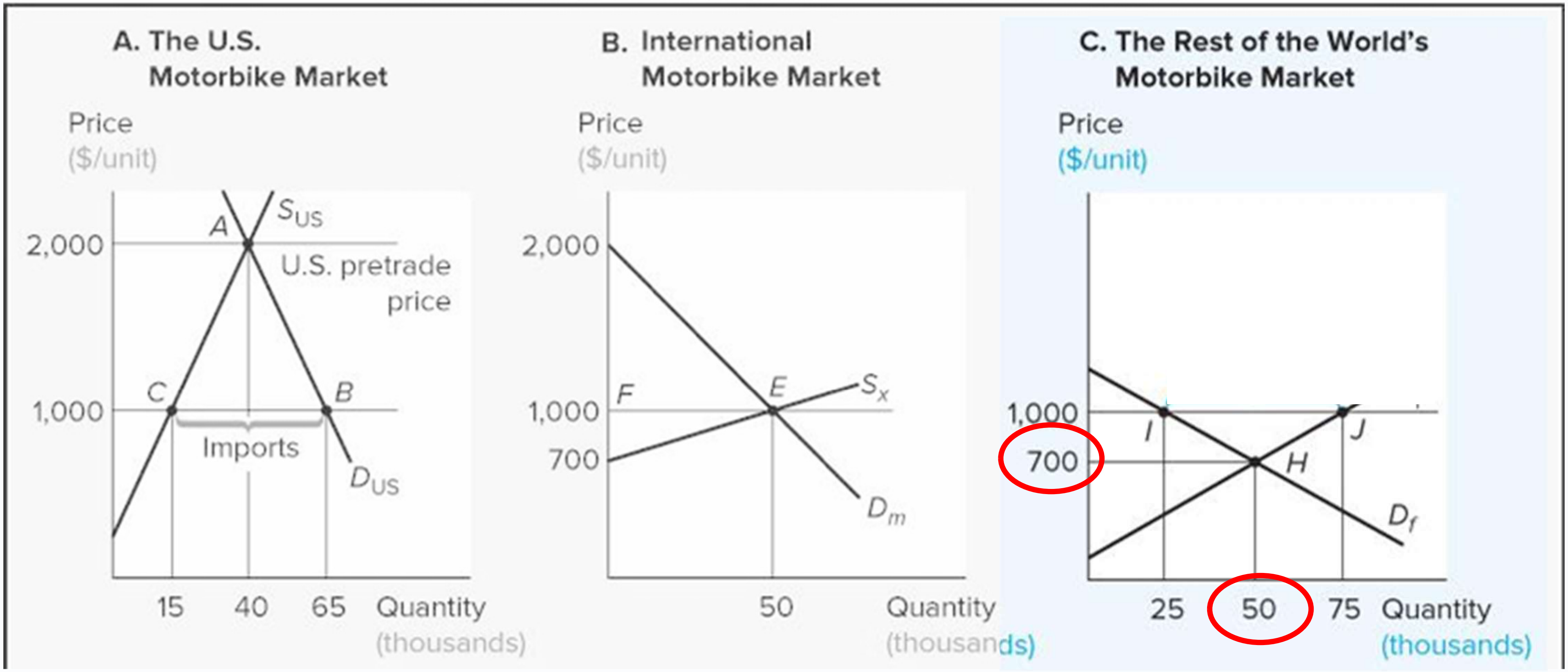

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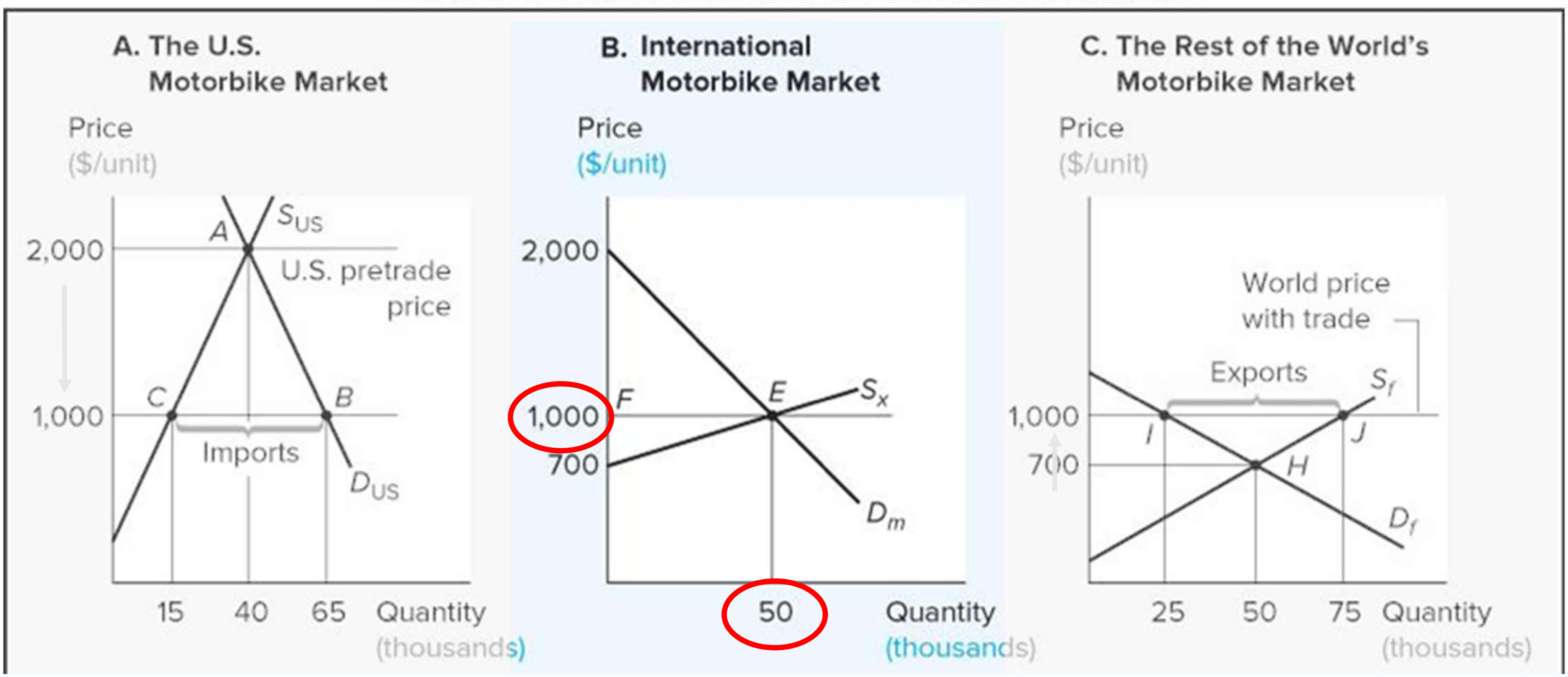

excess demand

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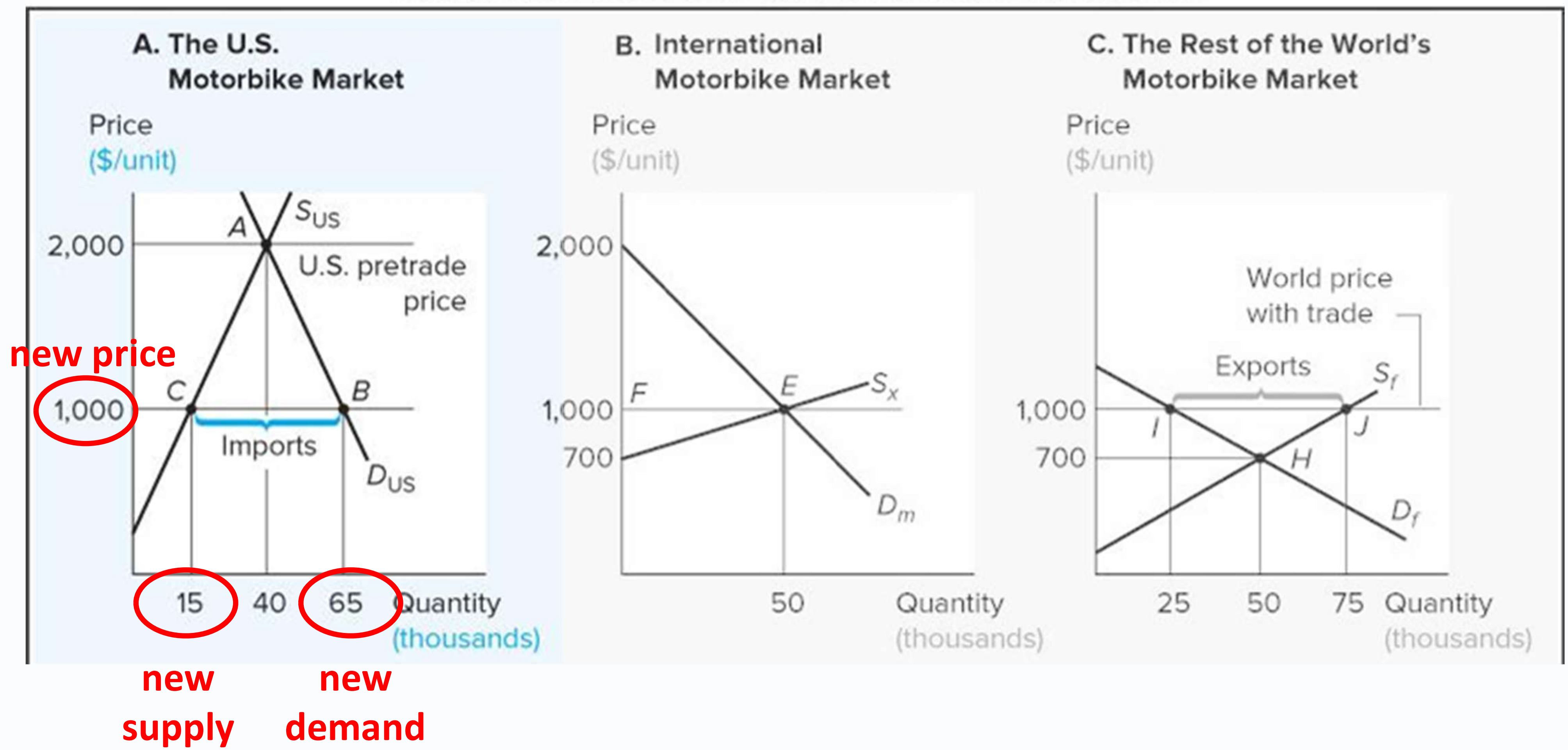

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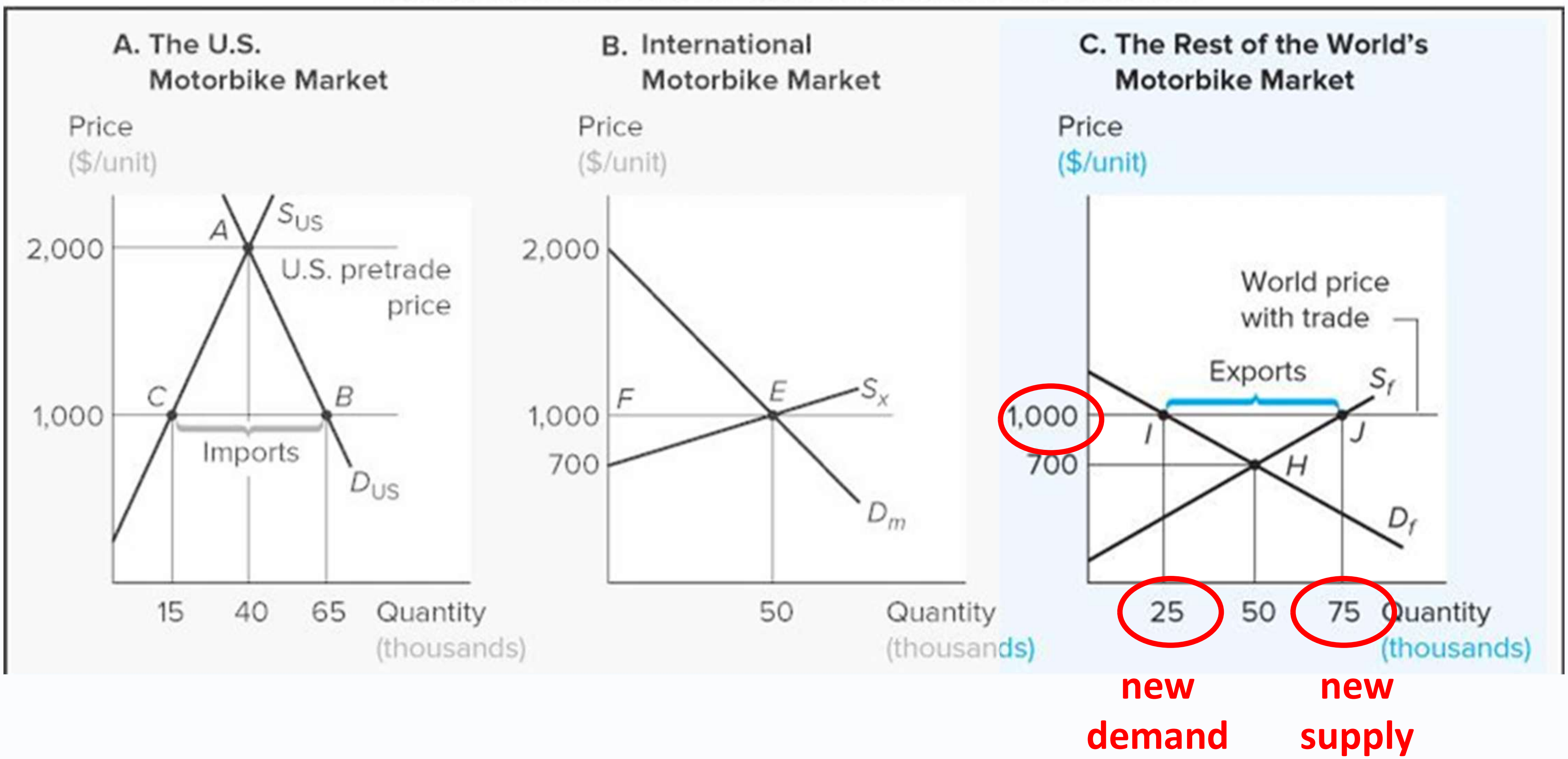

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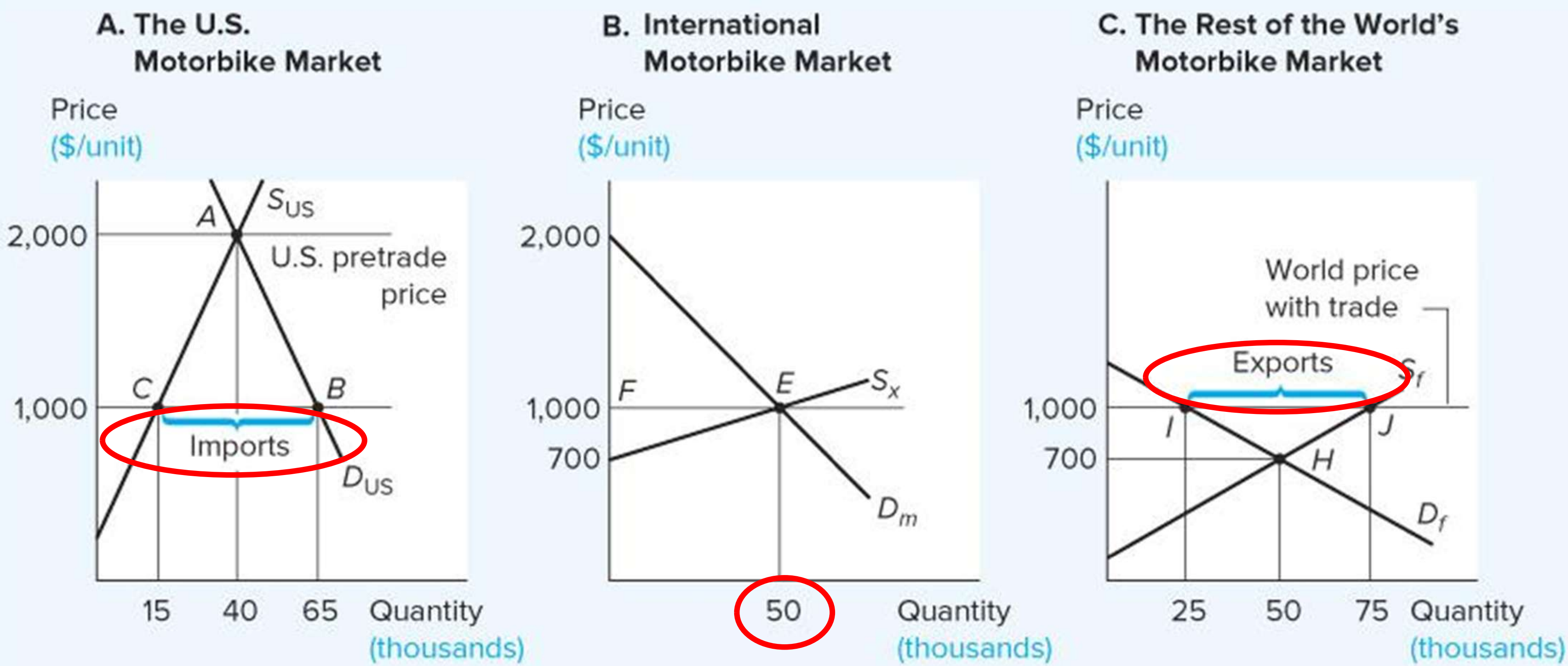


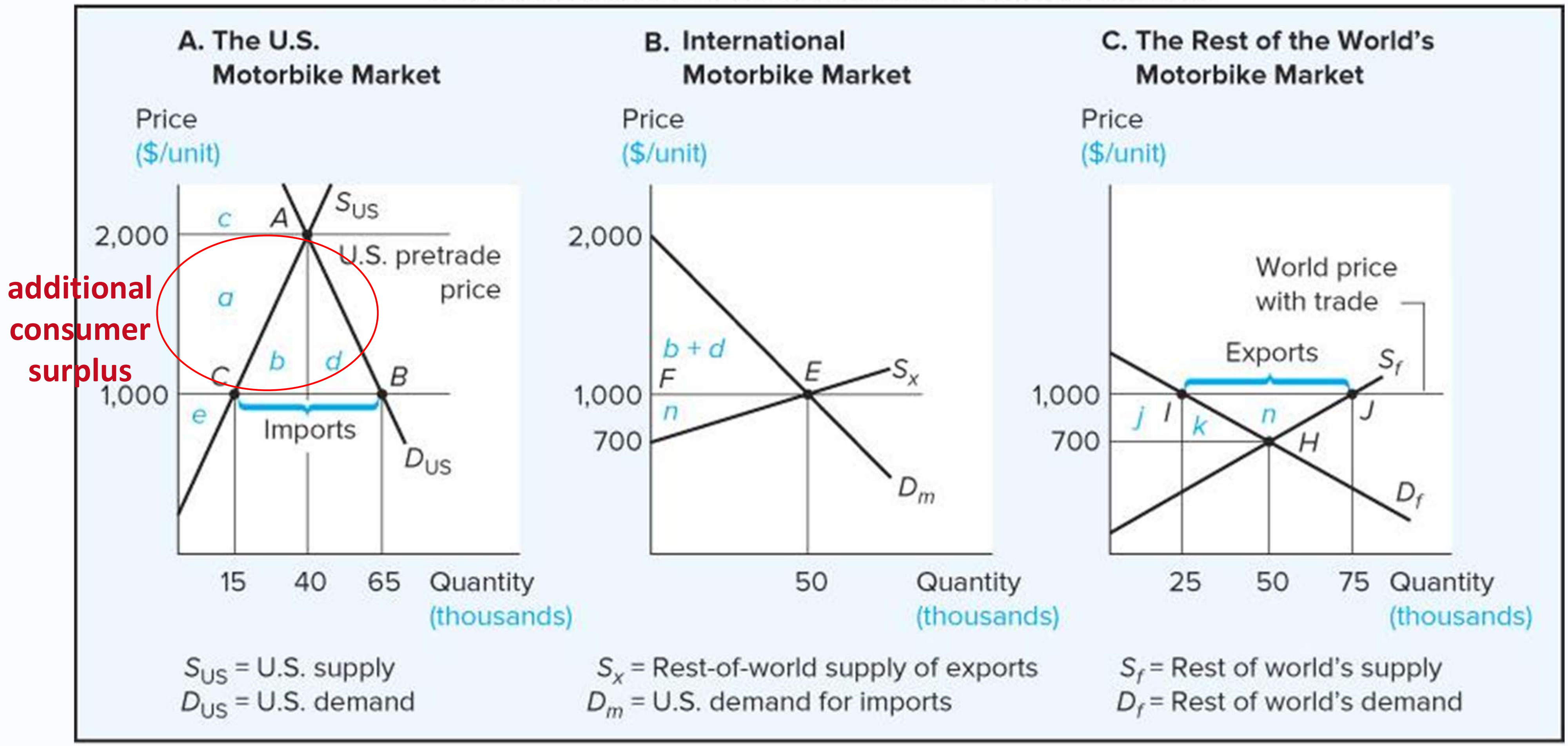
S_{US} = U.S. supply
D_{US} = U.S. demand

S_x = Rest-of-world supply of exports
($S_x = S_f - D_f$)
D_m = U.S. demand for imports
($D_m = D_{US} - S_{US}$)

S_f = Rest of world's supply
D_f = Rest of world's demand

The Effects of Trade on Well-Being of Producers, Consumers, and the Nation as a Whole

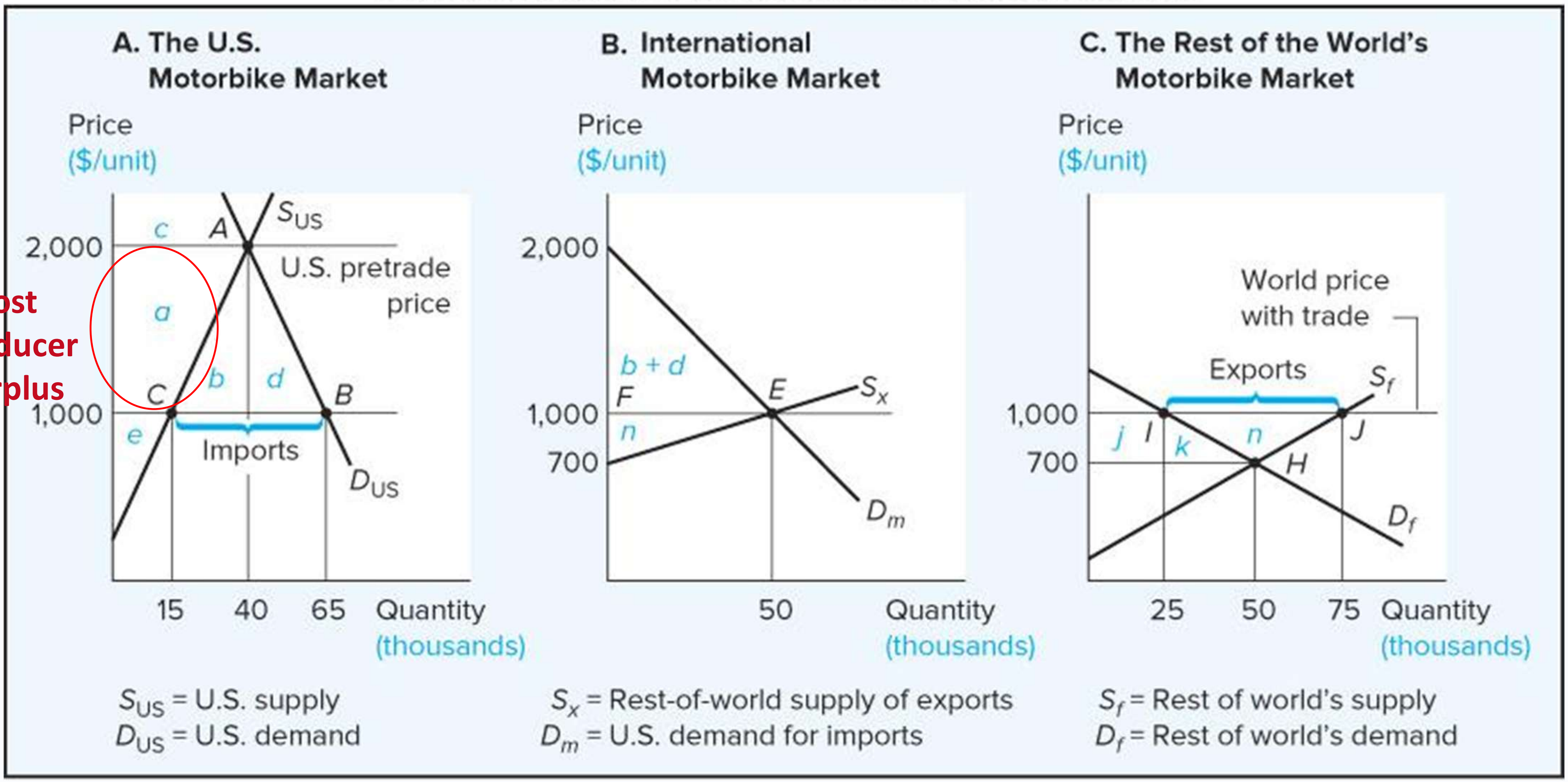
Additional consumer surplus = a + b + d



The Effects of Trade on Well-Being of Producers, Consumers, and the Nation as a Whole

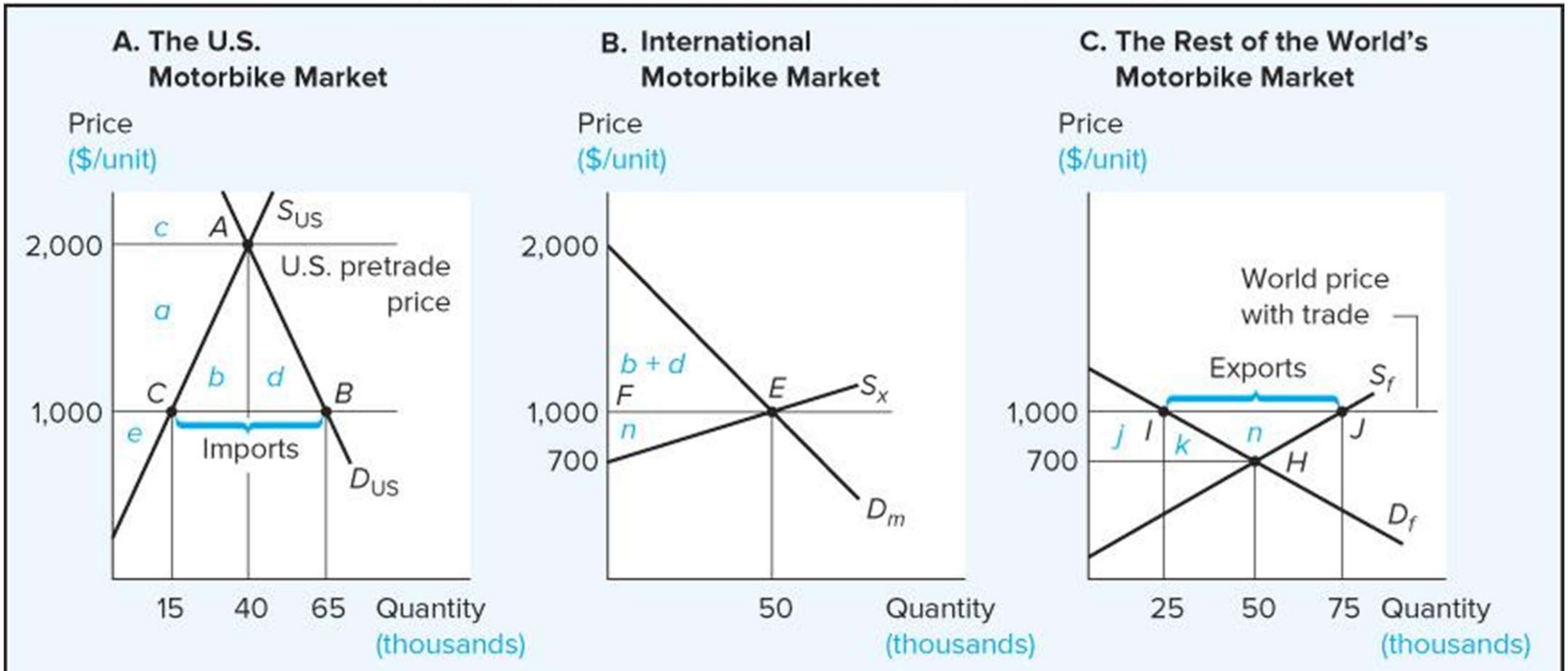
Lost producer surplus = -a

lost producer surplus



The Effects of Trade on Well-Being of Producers, Consumers, and the Nation as a Whole

Net US gain/loss = consumer surplus - producer surplus
= (a + b + d) – a = b + d



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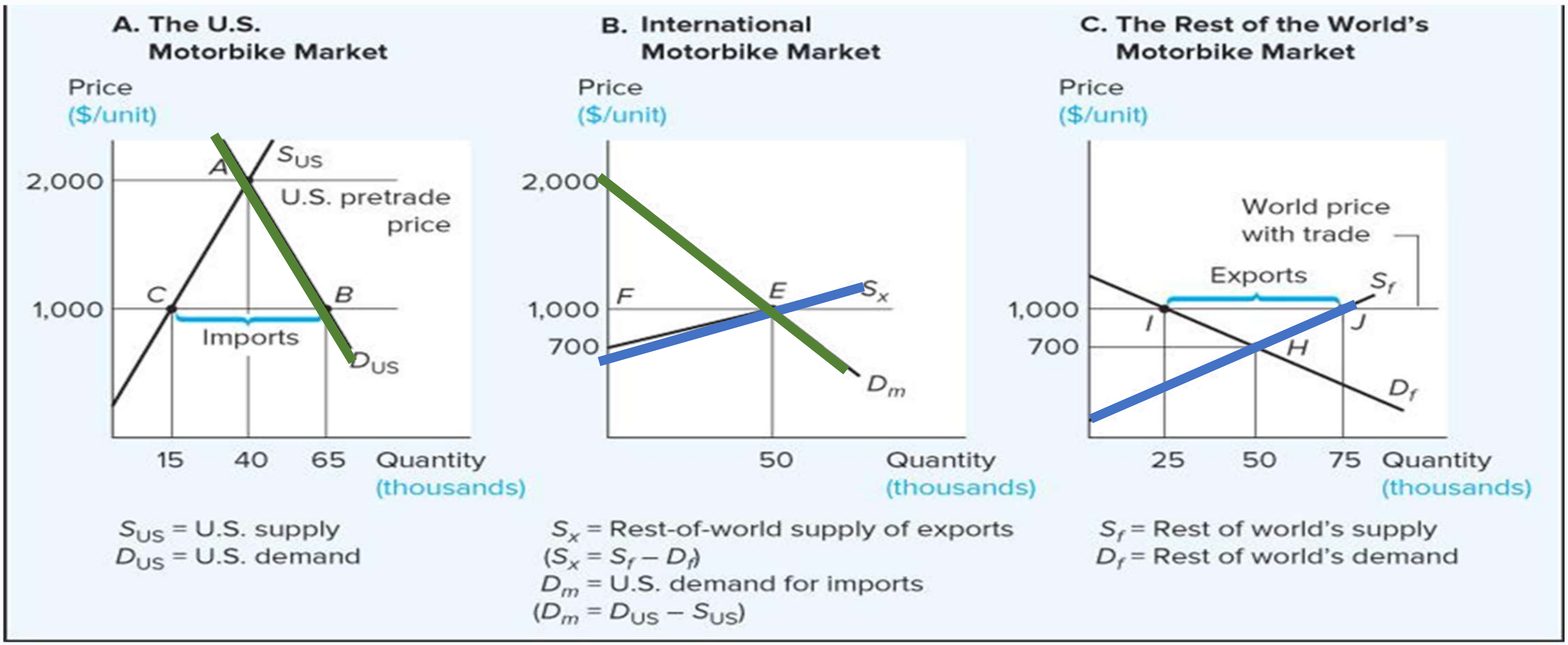
Which Country Gains More?

The side with the less elastic (steeper) trade curve (import demand curve or export supply curve) gains more.



Summary: Welfare Effects of Free Trade

United States (Importing Country)			
Group	**Surplus with Free Trade**	**Surplus with No Trade**	**Net Effect of Trade**
Consumers	a + b + c + d	c	a + b + d
Producers	e	a + e	− a [a loss]
U.S. as a whole (consumers + producers)	a + b + c + d + e	c + a + e	b + d

Rest of the World	
Group	**Net Effect of Trade**
Consumers	− (j + k) [a loss]
Producers	j + k + n
Rest of the world as a whole	n